NANO 2314 Quiz 07 — Review Packet

Practice for Quiz 07, Friday, October 23, 2026 · Fall 2026 · Prof./Coach Emori

18 problems on the ideas Quiz 07 covers, 17 of them free response: write your answer, with your work, in the space given. Try Part 1 first, then check yourself in Part 2. Multi-step answers are drawn step by step, and the **In words** box under each is the kind of written answer that earns full credit. No calculator needed.

Problem 1 is a question from Quiz 06, word for word; its answer shows how it was scored and the mistakes that cost the most points. The **Quiz 07 Practice Drill** (https://claude.ai/artifact/T7JJkS9pK8fc3Q2c4b1Gk3) gives fresh multiple-choice versions of all of these ideas, with new numbers each round.

Topic	Practices	Problems
1. Why Ae^(Xt) is a damped oscillation	split, the i, what each factor does, matching a plot	1–5
2. Fourier transform basics	time and frequency domains, what |X(f)| shows and hides	6–10
3. Frequencies from graphs	periods and frequencies off a graph, units, sketching a sine	11–15
4. Spectrum and waveform	sketching spectra, reading a spectrum back into a waveform	16–18

Part 1 · Problems

1. Why Ae^(Xt) is a damped oscillation

Problem 1 FREE RESPONSE Explain why Ae^(Xt) is a damped oscillation (from Quiz 06)

This problem is from Quiz 06, word for word. The damped-oscillator equation was $m\frac{d^2x}{dt^2} + 2m\gamma\frac{dx}{dt} + m\omega^2 x = 0$.

The differential equation above was transformed to the quadratic equation $X^2 + 2\gamma X + \omega^2 = 0$, one of whose two roots is

$$X = -\gamma + \sqrt{\gamma^2 - \omega^2}.$$

The solution $x(t)$ to the differential equation is then

$$x(t) = A\,e^{Xt}, \quad \text{where } A \text{ is a constant.}$$

Explain clearly how $x(t) = A\,e^{Xt}$, where $X = -\gamma + \sqrt{\gamma^2 - \omega^2}$, is a **damped oscillation** if γ is not zero but much smaller than ω ($\gamma \ll \omega$). Your answer should have $i = \sqrt{-1}$ somewhere.

Problem 2 FREE RESPONSE Split the exponential

Write $e^{\left(-\beta + \sqrt{\beta^2 - \omega^2}\right)t}$ as a **product** of two exponentials.

Problem 3 FREE RESPONSE Put the numbers back into Ae^(Xt)

A damped oscillator has $\gamma = 4\ \text{s}^{-1}$ and $\omega = 5\ \text{s}^{-1}$.

a. Find $X = -\gamma + \sqrt{\gamma^2 - \omega^2}$.

b. Write $x(t) = A\,e^{Xt}$ as a product of two factors.

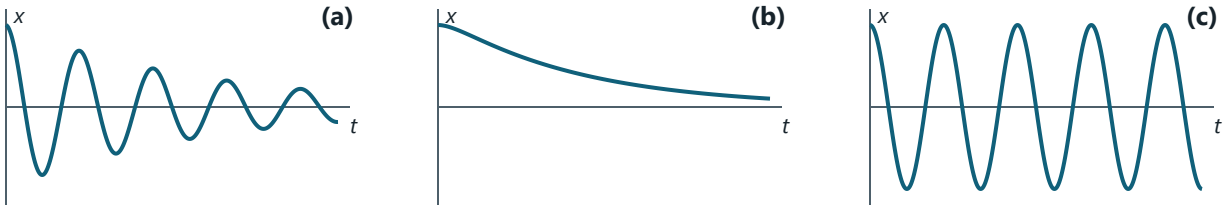
c. What does each factor do? At what angular frequency does x oscillate, and what makes its amplitude shrink?

Problem 4 FREE RESPONSE Find the error

A damped oscillator has the solution $x(t) = A\,e^{Xt}$ with $X = -\gamma + \sqrt{\gamma^2 - \omega^2}$, where $0 < \gamma \ll \omega$. A student writes: "Since $\gamma \ll \omega$, $\sqrt{\gamma^2 - \omega^2} \approx i\omega$, so $x(t) = A\,e^{i\omega t}$, which is a damped oscillation." What is wrong? Write the corrected $x(t)$ and say why it is a damped oscillation.

Problem 5 FREE RESPONSE Which γ matches each plot?

Someone claims $x(t) = A\,e^{Xt}$, with $X = -\gamma + \sqrt{\gamma^2 - \omega^2}$, $\gamma \geq 0$ and $\omega > 0$, matches each plot below (one at a time). For each plot, what must be true of γ ($\gamma = 0$, $0 < \gamma < \omega$, or $\gamma > \omega$), and does X contain an i?



2. Fourier transform basics

Problem 6 FREE RESPONSE Fill in the blanks

A cantilever's deflection recorded versus time, $x(t)$, is _____-domain data. Its Fourier transform $|X(f)|$ is a _____-domain spectrum, plotted against _____. Each harmonic gives a peak: the peak's **position** tells you the harmonic's _____, its **height** tells you the harmonic's _____, and its **width** tells you how quickly the harmonic _____.

Problem 7 FREE RESPONSE What can |X(f)| tell you?

For each item, say whether you can read it from a spectrum $|X(f)|$ like the ones in the homework (yes or no), with a one-line reason.

a. which frequencies the waveform contains

b. how large each harmonic is compared with the others

c. how the harmonics line up in time (their relative phase)

d. how quickly a harmonic dies away

e. the exact shape of the waveform, point by point

Problem 8 FREE RESPONSE Same harmonics, different phase

$x_1(t) = 4\ \text{nm}\sin(2\pi \cdot 100\ \text{Hz} \cdot t) + 1\ \text{nm}\sin(2\pi \cdot 300\ \text{Hz} \cdot t)$

$x_2(t) = 4\ \text{nm}\sin(2\pi \cdot 100\ \text{Hz} \cdot t) + 1\ \text{nm}\sin(2\pi \cdot 300\ \text{Hz} \cdot t + \pi)$

In x_2 the 300 Hz harmonic is shifted by half a cycle, so the two waveforms look different. (a) Sketch $|X(f)|$ for x_1, labeling the frequency axis. (b) How does $|X(f)|$ for x_2 compare? Explain.

Problem 9 FREE RESPONSE Damping and peak width

Oscillators A and B both ring at 400 Hz. A keeps ringing for hundreds of cycles; B dies away within about three cycles. Sketch both spectra $|X(f)|$ on the same axes and label which is which.

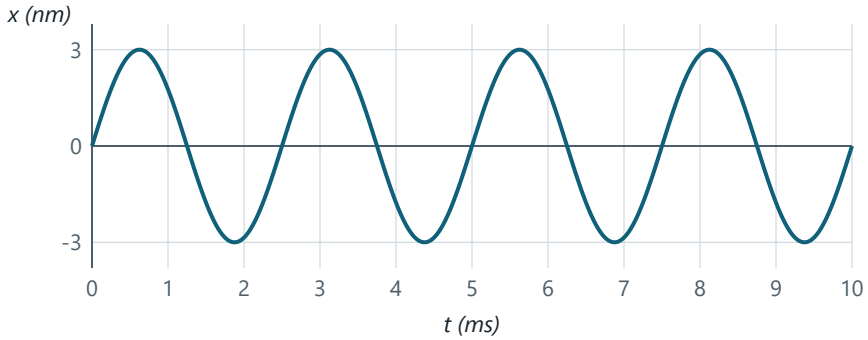
Problem 10 FREE RESPONSE Harmonics of a sung note

A sung note has its fundamental at 220 Hz. (a) At which frequencies are the next three harmonics (overtones)? (b) Sketch the spectrum of a pure 220 Hz sine wave, and a possible spectrum of the sung note, on separate axes.

3. Frequencies from graphs

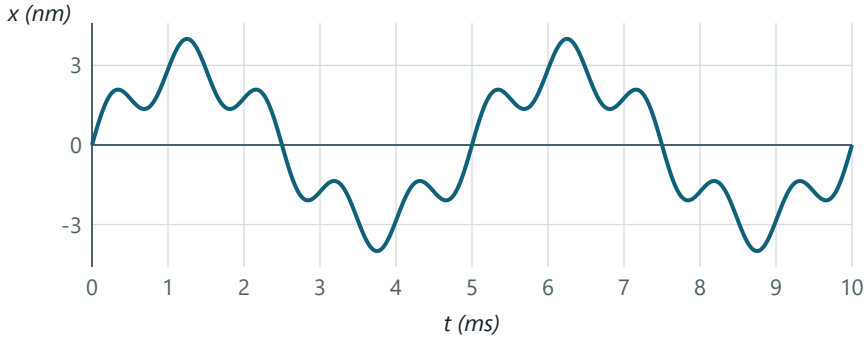
Problem 11 FREE RESPONSE Period and frequency from a graph

Find the period T and the frequency f of this signal. Show your work.



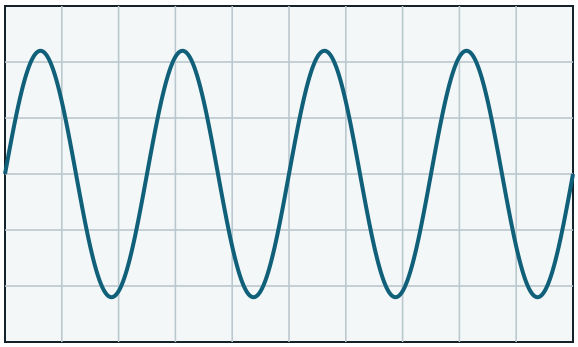
Problem 12 FREE RESPONSE Two harmonics from one graph

This signal is the sum of two harmonics: a large, slow one and a small, fast one. Estimate (i) the period and (ii) the frequency of each. Show your work, for example by marking the graph.



Problem 13 FREE RESPONSE Oscilloscope: time per division

The oscilloscope below is set to $0.5\ \text{ms}$ per division (one division is one grid square across). What is the frequency of the signal?

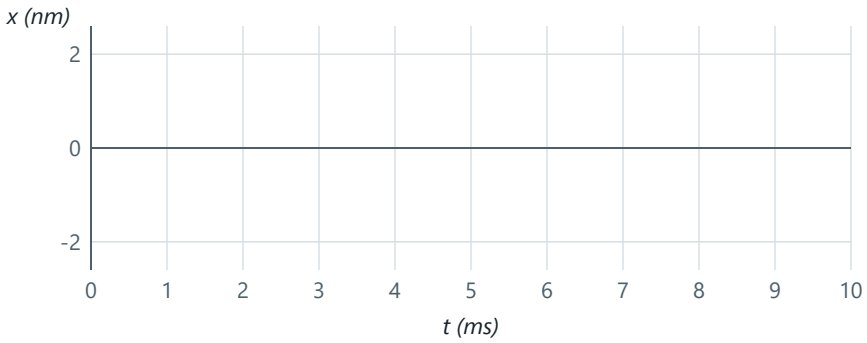


Problem 14 FREE RESPONSE Units: period, frequency, angular frequency

a. A signal repeats every $4\ \mu\text{s}$. What is its frequency?
b. A signal has frequency $2\ \text{kHz}$. What is its period?
c. A signal has frequency $250\ \text{Hz}$. What is its angular frequency ω?

Problem 15 FREE RESPONSE Sketch a sine from its frequency

On the axes below, sketch $x(t) = 2\ \text{nm}\sin(2\pi \cdot 400\ \text{Hz} \cdot t)$ from 0 to 10 ms.



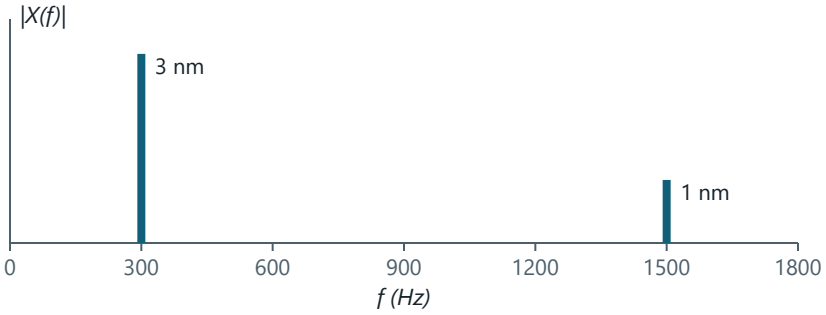
4. Spectrum and waveform

Problem 16 FREE RESPONSE Sketch the spectrum of a two-harmonic waveform

Sketch the Fourier transform $|X(f)|$ of the waveform in Problem 12. Label the horizontal axis, be as quantitative as you can about where the peaks are, and show their relative heights.

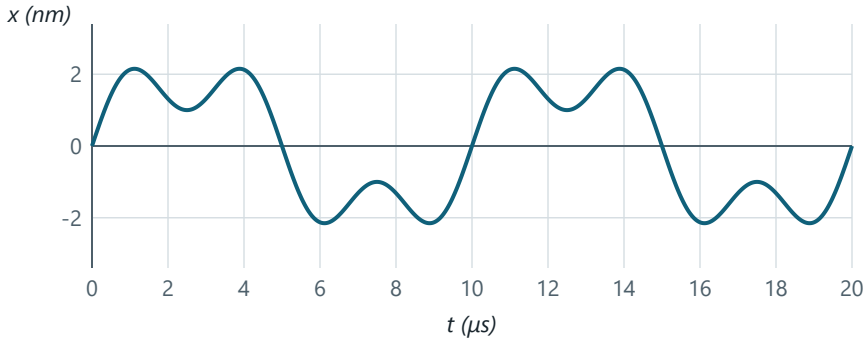
Problem 17 FREE RESPONSE From a spectrum back to the waveform

The spectrum below belongs to a signal made of sine waves that all start at zero (phase 0). Peak heights are amplitudes in nm. (a) Write $x(t)$. (b) Describe or sketch $x(t)$ from 0 to 10 ms: how many big cycles, and how many small ripples on each?

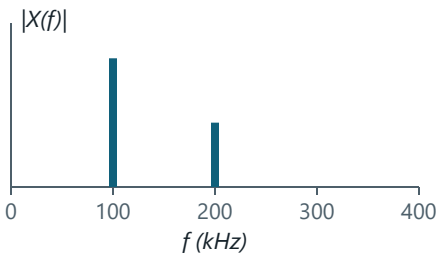


Problem 18 MULTIPLE CHOICE Pick the matching spectrum

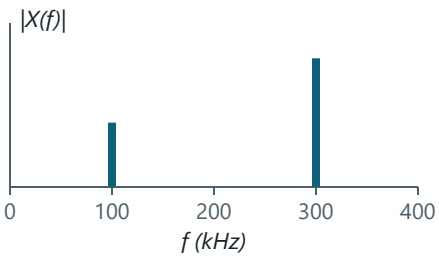
Which spectrum belongs to this waveform?



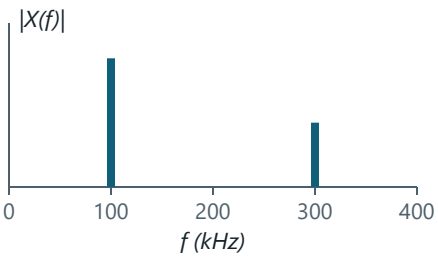
A.



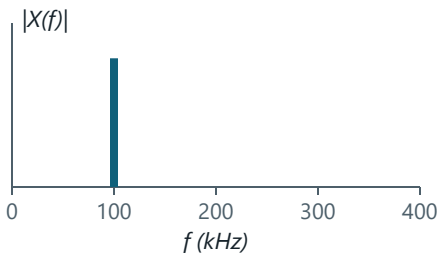
B.



C.



D.



Part 2 · Answers and explanations

1. Why Ae^(Xt) is a damped oscillation

Problem 1 Explain why Ae^(Xt) is a damped oscillation (from Quiz 06)

$$x(t) = A\,e^{Xt} = A\,e^{\left(-\gamma + \sqrt{\gamma^2 - \omega^2}\right)t}$$

↓ split the exponent: $e^{a+b} = e^a e^b$

$$x(t) = A\,e^{-\gamma t}\,e^{\sqrt{\gamma^2 - \omega^2}\,t}$$

↓ $\gamma \ll \omega$: pull out $\sqrt{-1} = i$

$$\sqrt{\gamma^2 - \omega^2} = i\sqrt{\omega^2 - \gamma^2} = i\,\omega_d \approx i\,\omega$$

Why: $\gamma^2 - \omega^2 < 0$, and $\sqrt{-1} = i$; with $\gamma \ll \omega$, $\omega_d \approx \omega$ (either is fine).

$A\,e^{-\gamma t}$ ↓ **decay:** real, shrinks because $\gamma > 0$

×

$e^{i\omega_d t}$ ↓ **oscillation:** size 1, repeats every $2\pi/\omega_d$

↓ multiply the two factors

Conclusion

an oscillation whose amplitude $A\,e^{-\gamma t}$ shrinks: a damped oscillation

In words. Splitting e^{Xt} gives $A\,e^{-\gamma t}$ times $e^{\sqrt{\gamma^2 - \omega^2}\,t}$. Because $\gamma \ll \omega$, the number under the root is negative, so $\sqrt{\gamma^2 - \omega^2} = i\sqrt{\omega^2 - \gamma^2} \approx i\omega$. The factor $e^{i\omega t}$ has size 1 and repeats, so it oscillates; the factor $e^{-\gamma t}$ shrinks, so the oscillation dies away.

How it was scored on Quiz 06 (1 point per step). (1) Split e^{Xt} into $e^{-\gamma t}$ times a second factor. (2) Show the i: $\gamma^2 - \omega^2 < 0$, so $\sqrt{\gamma^2 - \omega^2} = i\sqrt{\omega^2 - \gamma^2}$ (or $\approx i\omega$). (3) Say that the factor with i oscillates. Euler's formula is not required. (4) Say that $e^{-\gamma t}$ shrinks the amplitude, so the product is a damped oscillation.

Common mistakes

- Dropping $e^{-\gamma t}$: writing $\sqrt{\gamma^2 - \omega^2} \approx i\omega$ and then $x(t) = A\,e^{i\omega t}$. Only the root is approximated; the $-\gamma$ stays. Without $e^{-\gamma t}$ nothing shrinks. This was the most-missed step on Quiz 06.
- Stopping at $x(t) = A\,e^{-\gamma t}e^{i\omega_d t}$. The formula still has to be read: say which factor oscillates and which one makes it die away (two of the four points).
- Writing "so it is a damped oscillation" with no reason. A label is not an explanation.
- Finding X with numbers (for example $X = -4 + 3i$) and stopping. Put X back into $A\,e^{Xt}$ and say what each factor does (Problem 3).
- Writing $i\sqrt{\gamma^2 - \omega^2}$: the i comes out with the flipped, positive difference, $i\sqrt{\omega^2 - \gamma^2}$.
- Writing $e^{-\gamma t + \cdots}$ as a sum $e^{-\gamma t} + e^{\cdots}$. A sum in the exponent becomes a product.

Problem 2 Split the exponential

$$e^{\left(-\beta+\sqrt{\beta^2-\omega^2}\right)t} = e^{-\beta t}\, e^{\sqrt{\beta^2-\omega^2}\,t}$$

Distribute t, then use $e^{a+b} = e^a e^b$. The first factor will be the decay and the second the oscillation.

Common mistakes

- $e^{-\beta t} + e^{\sqrt{\beta^2-\omega^2}\,t}$: the exponential of a sum is a product, not a sum.
- $e^{-\beta t}\sqrt{\beta^2-\omega^2}\,t$: the root has to stay in an exponent.

Problem 3 Put the numbers back into Ae^(Xt)

$$X = -4+\sqrt{16-25} = -4+\sqrt{-9} = -4+3i\ \mathrm{s}^{-1}$$

↓ put X back into $A\,e^{Xt}$ and split

$$x(t) = A\,e^{(-4+3i)t} = A\,e^{-4t}\,e^{3it}$$

$A\,e^{-4t}$	×	e^{3it}
↓		↓
decay: the amplitude shrinks by a factor e^{-4} each second		**oscillation:** angular frequency 3 rad/s

In words. The real part of X, -4, gives the decaying factor e^{-4t}; the i part, 3, gives e^{3it}, which oscillates at 3 rad/s. So x oscillates at 3 rad/s while its amplitude dies away.

Common mistakes

- Stopping at $X = -4+3i$: finding X is only the start (this happened on Quiz 06).
- Saying it oscillates at 5 rad/s: that is ω. The motion oscillates at the i part of X, $\sqrt{25-16} = 3$ rad/s.
- Swapping the parts: $e^{-3t}e^{4it}$. The real part of X sets the decay; the i part sets the oscillation.

Problem 4 Find the error

The $-\gamma$ in X was dropped. Only the square root is approximated: $X = -\gamma+\sqrt{\gamma^2-\omega^2} \approx -\gamma+i\omega$, so $x(t) \approx A\,e^{-\gamma t}\,e^{i\omega t}$.

$A\,e^{i\omega t}$ alone has size $|A|$ forever: it oscillates but never dies away. The factor $e^{-\gamma t}$ makes the amplitude shrink, and that is the damping.

Common mistakes

- Saying "nothing is wrong": $|e^{i\omega t}| = 1$ for every t, so $A\,e^{i\omega t}$ is not damped.
- Blaming ω instead of ω_d: for $\gamma \ll \omega$ that is fine; the real error is the missing $e^{-\gamma t}$.

Problem 5 Which γ matches each plot?

a. Crests shrink while the curve keeps crossing zero: $0 < \gamma < \omega$ (weak damping). Then $\gamma^2 - \omega^2 < 0$, so X contains i (the oscillation), and $e^{-\gamma t}$ makes the crests shrink.

b. Creeps to zero without crossing it: $\gamma > \omega$ (strong damping). The root is real and smaller than γ, so X is a negative real number with no i: decay without oscillation.

c. Every crest reaches the same height: $\gamma = 0$. Then $X = \sqrt{-\omega^2} = i\omega$, and $e^{i\omega t}$ goes around forever without shrinking.

Common mistakes

- Saying (b) still has an i: with $\gamma > \omega$ the number under the root is positive, so no i and no oscillation.
- Calling (c) weakly damped: the crests do not shrink at all, so $\gamma = 0$.

2. Fourier transform basics

Problem 6 Fill in the blanks

time; frequency; frequency f (in Hz); frequency; amplitude (its size compared with the other harmonics); dies away (how strongly it is damped).

Common mistakes

- Swapping the domains: the waveform $x(t)$ is versus time (time domain); the spectrum is versus frequency (frequency domain).
- Reading height as frequency: frequency is where a peak sits along the axis; height is how big that harmonic is.

Problem 7 What can |X(f)| tell you?

a. Yes: one peak at each frequency present.

b. Yes: peak heights follow the amplitudes.

c. No: $|X(f)|$ throws the phase away; shifting a harmonic in time changes the waveform but not $|X(f)|$.

d. Yes: a harmonic that dies quickly gives a broad peak.

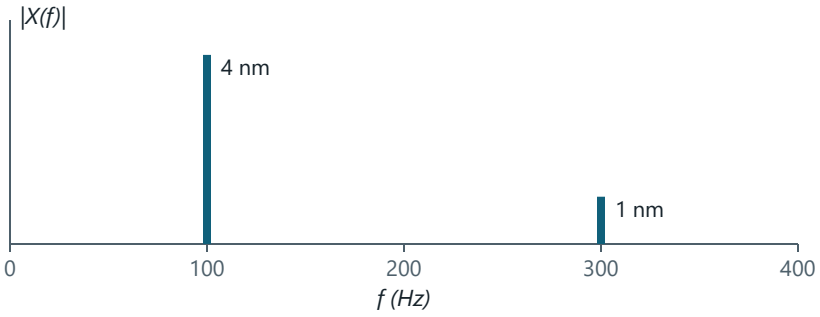
e. No: without the phases, different waveforms can have the same $|X(f)|$ (Problem 8), so the waveform cannot be rebuilt from it.

Common mistakes

- Saying yes to (c) or (e): the spectra we plotted are magnitudes, which carry no phase information.

Problem 8 Same harmonics, different phase

a. Two peaks: at 100 Hz (tall, amplitude 4 nm) and at 300 Hz (a quarter as tall, 1 nm), and nothing else.



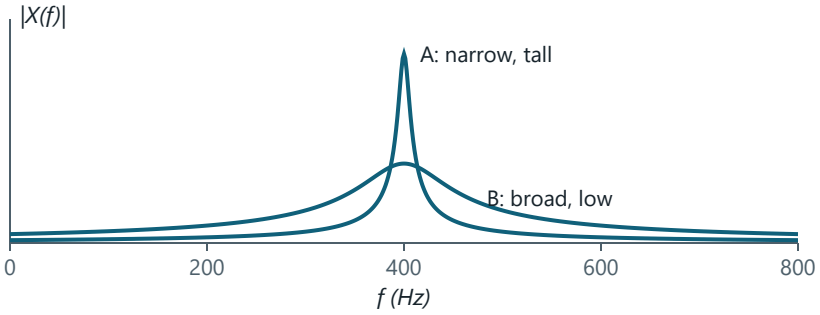
b. Identical. The same two harmonics with the same amplitudes are present; only the timing (phase) of one differs, and $|X(f)|$ does not record phase.

Common mistakes

- Drawing the 300 Hz peak of x_2 pointing down: $|X(f)|$ is a magnitude, never negative.
- Moving or removing the 300 Hz peak: a phase shift does not change how often the harmonic repeats or how big it is.

Problem 9 Damping and peak width

Both peaks sit at 400 Hz, the frequency of the oscillation. A, which lasts a long time, gives a **narrow** peak; B, which dies quickly, gives a **broad** (and lower) peak. Faster decay, broader peak.



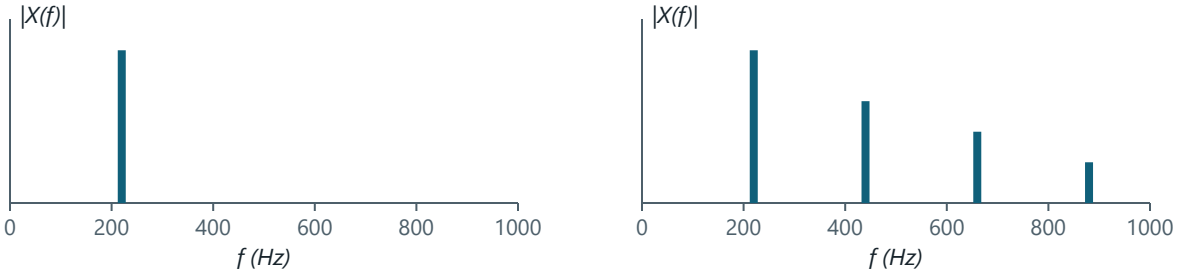
Common mistakes

- Putting the two peaks at different frequencies: both oscillate at 400 Hz.
- Making the strongly damped one narrow: it is the other way round.

Problem 10 Harmonics of a sung note

a. Whole-number multiples of the fundamental: $2 \times 220 = 440$ Hz, $3 \times 220 = 660$ Hz and $4 \times 220 = 880$ Hz.

b. A pure sine has one frequency, so one peak at 220 Hz. The sung note has a row of peaks evenly spaced by 220 Hz; their heights depend on the voice.



Common mistakes

- Steps of a fixed size (270, 320, 370 Hz): harmonics are multiples, not equal additions.
- Drawing harmonics for the pure sine: a pure sine has exactly one peak.

3. Frequencies from graphs

Problem 11 Period and frequency from a graph

4 full cycles take 10 ms

↓ divide the time by the number of cycles

$$T = \frac{10\ \text{ms}}{4} = 2.5\ \text{ms}$$

↓ frequency is one over the period

$$f = \frac{1}{T} = \frac{1}{2.5\ \text{ms}} = 0.4\ \text{kHz} = 400\ \text{Hz}$$

Why: one over a millisecond is 1000 per second, a kilohertz.

In words. Four cycles take 10 ms, so one takes 2.5 ms; the frequency is one over that, 400 Hz.

Common mistakes

- $f = 0.4$ Hz: $1/(2.5\ \text{ms})$ is 0.4 per millisecond, which is 400 per second.
- Measuring crest to trough as a period: that is half a cycle and gives twice the frequency.

Problem 12 Two harmonics from one graph

$$T_1 = \frac{10\ \text{ms}}{2} = 5\ \text{ms}$$

Why: slow harmonic, big crest to big crest: 2 cycles in 10 ms.

↓ count the small ripples in one slow cycle

$$T_2 = \frac{T_1}{5} = 1\ \text{ms}$$

Why: fast harmonic: 5 small ripples ride on each slow cycle.

↓ frequency is one over the period

$$f_1 = \frac{1}{5\ \text{ms}} = 200\ \text{Hz}, \qquad f_2 = \frac{1}{1\ \text{ms}} = 1000\ \text{Hz}$$

In words. The big wave repeats every 5 ms, so $f_1 = 200$ Hz. Five small ripples ride on each big cycle, so the fast harmonic repeats five times as often: $T_2 = 1$ ms and $f_2 = 1000$ Hz.

The fast harmonic is hard to read exactly by eye; on the quiz, an estimate within about 25% earns full credit, as long as your work shows how you got it.

Common mistakes

- Counting only every other ripple (or both the crest and the trough of each ripple): that halves or doubles f_2.
- Reading the slow period from crest to trough: that is half a cycle.

Problem 13 Oscilloscope: time per division

4 cycles across 10 divisions, so one period spans 2.5 divisions

↓ times the time per division

$$T = 2.5 \times 0.5\ \text{ms} = 1.25\ \text{ms}$$

↓ frequency is one over the period

$$f = \frac{1}{1.25\ \text{ms}} = 0.8\ \text{kHz} = 800\ \text{Hz}$$

In words. One period is 2.5 divisions, which at 0.5 ms per division is 1.25 ms; the frequency is one over that, 800 Hz.

Common mistakes

- Using the number of divisions per period as the period: multiply by the time per division first.
- $1/(0.5\ \text{ms}) = 2000$ Hz: that is one over a single division, not one over a whole period.

Problem 14 Units: period, frequency, angular frequency

a. $f = 1/(4\ \mu\text{s}) = 0.25\ \text{MHz} = 250\ \text{kHz}$ (one over a microsecond is a megahertz).

b. $T = 1/(2\ \text{kHz}) = 0.5\ \text{ms}$ (one over a kilohertz is a millisecond).

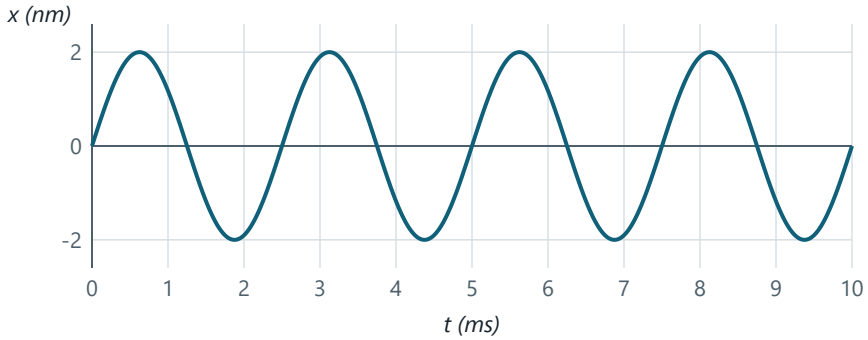
c. $\omega = 2\pi f = 2\pi \times 250\ \text{Hz} = 500\pi\ \text{rad/s} \approx 1570\ \text{rad/s}$.

Common mistakes

- Prefixes off by 1000: $1/\mu\text{s}$ is MHz and $1/\text{ms}$ is kHz.
- $\omega = 250$ rad/s or $\omega = 250\pi$ rad/s: one cycle is 2π radians, so $\omega = 2\pi f$.

Problem 15 Sketch a sine from its frequency

$T = 1/(400\ \text{Hz}) = 2.5\ \text{ms}$, so the sine completes 4 cycles in 10 ms: it starts at zero, rises to 2 nm at 0.625 ms, and crosses zero going up at 2.5, 5 and 7.5 ms.



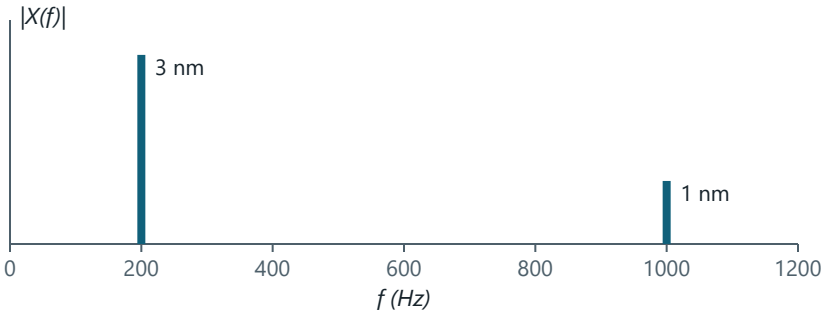
Common mistakes

- Drawing 400 cycles, or 0.4 cycles: convert first, $1/(400\ \text{Hz}) = 2.5\ \text{ms}$.
- Starting at a crest: $\sin 0 = 0$, so the sine starts at zero and rises.

4. Spectrum and waveform

Problem 16 Sketch the spectrum of a two-harmonic waveform

Two peaks, one per harmonic: at $f_1 = 200$ Hz (tall, the 3 nm harmonic) and at $f_2 = 1000$ Hz (about a third as tall, the 1 nm ripple). Nothing else.



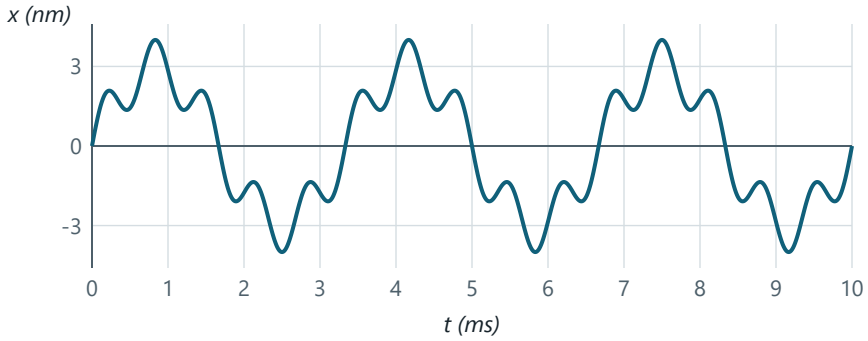
Common mistakes

- Peaks at 5 and 1 (the periods in ms): the axis is frequency, so use $f = 1/T$.
- Equal heights, or the taller peak at 1000 Hz: the slow harmonic is the big one, so its peak is taller.
- Extra peaks at 400, 600, 800 Hz: only two harmonics are present.

Problem 17 From a spectrum back to the waveform

a. $x(t) = 3\ \text{nm}\sin(2\pi \cdot 300\ \text{Hz} \cdot t) + 1\ \text{nm}\sin(2\pi \cdot 1500\ \text{Hz} \cdot t)$.

b. The 300 Hz harmonic has period $1/(300\ \text{Hz}) \approx 3.3$ ms, so 3 big cycles in 10 ms. The 1500 Hz harmonic is 5 times as fast, so 5 small ripples ride on each big cycle (15 in 10 ms), with about a third of the size.



Common mistakes

- Writing $\sin(300t)$: inside the sine goes $2\pi ft$; without 2π the number is ω, not f.
- Swapping the amplitudes: the taller peak is the bigger, slower harmonic.

Problem 18 Pick the matching spectrum

C. The big wave makes 2 cycles in 20 µs (period 10 µs, so 100 kHz) and carries 3 ripples per cycle (300 kHz). The big one is about twice the size of the ripple, so the 100 kHz peak is twice as tall.

- If you chose **A**: its second peak is at 200 kHz, but there are 3 ripples per big cycle, so the ripple is at 300 kHz.
- If you chose **B**: its heights are swapped: the slow harmonic is the large one, so its peak is taller.
- If you chose **D**: it has only one peak, but the ripples are a second harmonic.